

Zaid AAAI Alignment track

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0.1 Background

Calibration and explainability are commonly treated as two independent requirements for trustworthy AI. Predictive calibration ensures that a model's confidence reflects its empirical likelihood of correctness, whereas post hoc explanations aim to provide human-understandable reasons for a model's decision. However, in high-stakes domains such as medical AI, users often rely not only on the predicted label and confidence score, but also on the certainty conveyed by the accompanying explanation.

0.2 Predictive Calibration and Post-hoc Explanations in Medical AI

In medical AI, reliable prediction requires more than high classification accuracy. A model should not only produce confidence scores that accurately reflect the likelihood of being correct, but should also provide explanations that meaningfully identify the image regions or features supporting its decision. These two aspects are commonly studied independently: **predictive calibration** concerns the reliability of a model's confidence estimates, whereas **post-hoc explainability** aims to reveal the evidence underlying a prediction through methods such as Grad-CAM, LIME and SHAP.

Emerging empirical evidence, however, suggests that these properties may be related. For example, MDCA (our CVPR paper) introduces a train-time calibration objective that explicitly encourages alignment between predicted confidence and empirical class frequencies. Interestingly, models trained with this calibration objective were also observed to produce more meaningful Grad-CAM explanations, suggesting that improving predictive calibration may influence the internal representations and decision boundaries used by the classifier. This raises an important question:

Research Insight

Research Question: Does improved predictive calibration systematically lead to more reliable post-hoc explanations, or is the observed relationship between calibration and explanation quality merely empirical?

This question is particularly relevant in medical imaging, where both confidence and explanation quality have direct implications for clinical trust. A highly confident diagnosis accompanied by an explanation highlighting clinically irrelevant regions may be unsafe, while a plausible saliency map associated with an unreliable probability estimate can similarly create a misleading sense of certainty. Ideally, the model’s *predictive reliability* and *explanatory reliability* should therefore be consistent: when a model is confident and correct, its explanation should reliably identify evidence responsible for that prediction, whereas uncertain or erroneous predictions should not be accompanied by deceptively confident explanations.

Motivated by this observation, we investigate whether there exists a direct mathematical relationship between predictive calibration and explanation calibration. Rather than treating confidence calibration and post-hoc explanation quality as independent properties, we seek to characterize conditions under which calibration of the predictive distribution constrains, correlates with, or improves the reliability of the resulting explanations. Establishing such a connection could provide a principled foundation for jointly reasoning about what a medical AI system predicts, how certain it is, and why it makes that prediction.

0.3 Calibration Preliminaries

Let an LLM-based decision-support model be denoted by f_θ . Given an input clinical case

$$x = (q, c),$$

where q denotes the user query and c denotes the clinical context, the model produces a prediction

$$\hat{y} = f_\theta(x),$$

together with a predictive confidence score

$$p_\theta(\hat{y} \mid x) \in [0, 1].$$

Here, $p_\theta(\hat{y} \mid x)$ represents the model’s estimated probability that the predicted answer $\hat{y}$ is correct for input x .

For an LLM, the input x may consist of a natural-language clinical question, patient symptoms, laboratory findings, radiology reports, and electronic health record notes. The output may consist of a predicted diagnosis, a confidence estimate, and a natural-language explanation. Thus, the full output of the system can be written as

$$F(x) = (\hat{y}, p_\theta(\hat{y} \mid x), e),$$

where e denotes the post hoc explanation generated for the prediction $\hat{y}$.

Classical predictive calibration requires that, among all examples for which the model assigns confidence c , the model is correct approximately c of the time. Formally, a model is calibrated if

$$\mathbb{P}(Y = \hat{Y} \mid p_\theta(\hat{Y} \mid X) = c) = c,$$

where Y denotes the true label and $\hat{Y}$ denotes the model prediction.

0.4 Mathematical Link Between Calibration and Post Hoc Explanations

However, this condition only calibrates the prediction. It does not ensure that the explanation communicates the same degree of uncertainty as the predictive model. To capture this, we define an explanatory certainty function

$$S(e, x, \hat{y}) \in [0, 1],$$

where $S(e, x, \hat{y})$ measures the degree of certainty conveyed by the explanation e for prediction $\hat{y}$ on input x . A high value of S indicates that the explanation presents the prediction as highly certain, while a low value indicates that the explanation communicates ambiguity or uncertainty.

For LLM-generated explanations, $S(e, x, \hat{y})$ may be estimated using linguistic markers of certainty and uncertainty, explanation stability across multiple generations, human-rated perceived certainty, or an auxiliary explanation-certainty model. For example, explanations containing phrases such as “clearly indicates”, “strongly confirms”, or “definitively shows” may receive higher certainty scores, whereas explanations containing phrases such as “may suggest”, “possibly indicates”, or “evidence is inconclusive” may receive lower certainty scores.

We define post hoc explanation generation as

$$e = h_\psi(x, \hat{y}, f_\theta),$$

where h_ψ is a post hoc explanation function that generates an explanation for the prediction of model f_θ . Explanation calibration requires that the certainty communicated by the explanation aligns with the predictive confidence of the model. **Formally, an explanation is calibrated if**

$$\mathbb{E} \left[S(h_\psi(X, \hat{Y}, f_\theta), X, \hat{Y}) \mid p_\theta(\hat{Y} \mid X) = c \right] = c.$$

This condition states that, for all cases where the model confidence is c , the explanation should also convey approximately c level of certainty.

Equivalently, for an individual input x , explanation calibration requires

$$S(e, x, \hat{y}) \approx p_\theta(\hat{y} \mid x).$$

Thus, the mathematical link between calibration and post hoc explanation can be expressed as

$$\text{Explanatory certainty} \approx \text{Predictive confidence}$$

or mathematically,

$$S(h_\psi(x, \hat{y}, f_\theta), x, \hat{y}) \approx p_\theta(\hat{y} \mid x).$$

To quantify deviations from this alignment, we define the Explanation Calibration Error as

$$\text{ExCE} = \mathbb{E}_{(x,y)} [|S(e, x, \hat{y}) - p_{\theta}(\hat{y} | x)|].$$

For a finite dataset of n examples, this can be estimated as

$$\widehat{\text{ExCE}} = \frac{1}{n} \sum_{i=1}^n |S(e_i, x_i, \hat{y}_i) - p_{\theta}(\hat{y}_i | x_i)|.$$

Here, each example consists of an input x_i , a prediction $\hat{y}_i$, a predictive confidence score $p_{\theta}(\hat{y}_i | x_i)$, and an explanation e_i .

We can further define a signed explanation calibration gap as

$$\Delta_i = S(e_i, x_i, \hat{y}_i) - p_{\theta}(\hat{y}_i | x_i).$$

If $\Delta_i > 0$, the explanation is overconfident because it conveys greater certainty than the model’s predictive confidence. If $\Delta_i < 0$, the explanation is underconfident because it conveys less certainty than the model’s predictive confidence. If $\Delta_i \approx 0$, the explanation is calibrated with respect to the model’s uncertainty.

This gives two directional forms of explanation miscalibration:

$$\text{OverExCE} = \frac{1}{n} \sum_{i=1}^n \max(0, \Delta_i),$$

and

$$\text{UnderExCE} = \frac{1}{n} \sum_{i=1}^n \max(0, -\Delta_i).$$

These quantities distinguish whether explanations systematically overstate or understate the model’s confidence.

0.5 Example: LLM-Based Medical AI

Consider an LLM-based medical decision-support system that receives the input

$$x = (q, c),$$

where

$$q = \text{“Does this patient have pneumonia?”}$$

and c contains the patient’s symptoms, blood test values, and radiology report. Suppose the LLM predicts

$$\hat{y} = \text{“Pneumonia likely”}$$

with predictive confidence

$$p_{\theta}(\hat{y} | x) = 0.55.$$

This means that the model is only moderately confident in the diagnosis.

Now consider the following post hoc explanation:

The diagnosis is supported by fever, cough, elevated white blood cell count, and visible opacity in the lung field, indicating pneumonia.

Although the predictive confidence is only 0.55, this explanation may sound highly certain to a clinician or patient because it presents the evidence in a definitive manner. Suppose the explanatory certainty function assigns

$$S(e, x, \hat{y}) = 0.90.$$

Then the explanation calibration gap is

$$\Delta = 0.90 - 0.55 = 0.35,$$

and the explanation calibration error is

$$\text{ExCE} = |0.90 - 0.55| = 0.35.$$

This indicates that the explanation substantially overstates the model’s confidence. The prediction communicates moderate uncertainty, whereas the explanation communicates high certainty.

A better calibrated explanation would be:

The findings are suggestive of pneumonia because the patient has fever, cough, elevated white blood cell count, and lung opacity. However, the evidence is not definitive, and other causes of these symptoms should also be considered.

For this explanation, suppose

$$S(e, x, \hat{y}) = 0.58.$$

Then

$$\text{ExCE} = |0.58 - 0.55| = 0.03.$$

This explanation is better calibrated because the certainty conveyed by the explanation is closely aligned with the model’s predictive confidence.

Thus, in LLM-based medical AI, trustworthiness requires not only calibrated prediction probabilities, but also calibrated explanations. A reliable system should satisfy both

$$p_\theta(\hat{y} \mid x) \approx \mathbb{P}(Y = \hat{Y} \mid p_\theta(\hat{Y} \mid X))$$

and

$$S(e, x, \hat{y}) \approx p_\theta(\hat{y} \mid x).$$

The first condition ensures that the model’s confidence is statistically meaningful, while the second ensures that the explanation faithfully communicates the model’s uncertainty to the user.